HastaDAQ strain-gauge board

Net labels connect everything: same name = same net. Companion to the main board (kicad/hastadaq). Reusable: any I2C master with 5 V and 3.3 V logic can drive it.

POWER: bulk + decoupling on the loom 5 V (any ferrite bead is part of the loom, not the board), the LDO makes the quiet 3.3 V that powers BOTH ADCs and the bridge excitation

J1
LOOM from main board (4-wire I2C): GND, +5V, SDA, SCL

GND 1 GND
+5V 2 5V
SDA 3 SDA
SCL 4 SCL

C1 47uF 16V (RS PRO LHK 5x7)
C2 100nF 50V X7R (Vishay K104)

U1
3.3 V LDO: MCP1700-3302E/TO (TO-92: pin 1 GND, 2 VIN, 3 VOUT)

+5V 2 VIN
GND 1 GND
VOUT 3 +3V3

C3 10uF 10V tantalum (RS PRO TCB X7R 1206)
C4 100nF 50V X7R (Vishay K104)

#FLG1 PWR_FLAG (+5V)
#FLG2 PWR_FLAG (GND)

Pin 2 is +5V, NOT 3V3: use a keyed plug, never a Qwiic cable.

Load is about 20 mA (bridges 2 x 4.7 mA + ADCs): the LDO dissipates about 35 mW (250 mA rating).

TEST POINTS: GND (scope clip) and the regulated +3V3

TP1 TP GND
TP2 TP +3V3

J2
LOOM OUT (pass-through, paralleled with J1): GND, +5V, SDA, SCL

GND 1 GND
+5V 2 5V
SDA 3 SDA
SCL 4 SCL

Second 1x4 header wired straight to the loom: daisy-chains another I2C board (+5V, 3.3 V logic).

U2
Adafruit ADS122C04 #1 (I2C 0x40)

+3V3 1 VIN
GND 2 GND
SDA 3 SDA
SCL 4 SCL
5 RST
6 DRDY
A0 7 ADS1_A0
A1 8 ADS1_A1
A2 9
A3 10 SCL
Ref+ 11
Ref- 12

ADS1 input filter (optional: 1k links + 100nF)

ADS1_SIG
ADS_REF
ADS1_A0

R1 0R link (ADS1_SIG - ADS1_A0)
R2 0R link (ADS_REF - ADS1_A1)
C5 100nF 50V X7R (DNP) (ADS1_A0 - ADS1_A1)

J3
HB1 half-bridge #1 (on beam): V, SIG, GND

+3V3 1 V
ADS1_SIG 2 SIG
GND 3 GND

Shared completion resistors (fixed) -> A1 of BOTH ADCs

+3V3
R5 350R fixed
ADS_REF
R6 350R fixed
GND

U3
Adafruit ADS122C04 #2 (address jumper bridged)

+3V3 1 VIN
GND 2 GND
SDA 3 SDA
SCL 4 SCL
5 RST
6 DRDY
A0 7 ADS2_A0
A1 8 ADS2_A1
A2 9
A3 10 SCL
Ref+ 11
Ref- 12

ADS2 input filter (optional: 1k links + 100nF)

ADS2_SIG
ADS_REF
ADS2_A0

R3 0R link (ADS2_SIG - ADS2_A0)
R4 0R link (ADS_REF - ADS2_A1)
C6 100nF 50V X7R (DNP) (ADS2_A0 - ADS2_A1)

J4
HB2 half-bridge #2 (on beam): V, SIG, GND

+3V3 1 V
ADS2_SIG 2 SIG
GND 3 GND

NOTES
1. Each half-bridge head (2 gauges) plugs into a 3-pin header: V (+3V3), SIG (mid-point), GND. The two fixed 350R completion resistors are shared; their mid-point (ADS_REF) goes to A1 of both ADCs. Excitation is the ADC's own VIN, so VREF_SUPPLY is ratiometric: no Ref+/Ref- wiring.
2. 3V3 comes ONLY from the on-board LDO (from the loom's 5 V): cable voltage drop and the main board's switching noise do not reach the bridges or the ADC reference. The ADC logic lines (SDA/SCL) are pulled up by the ADS122C04 breakouts to this 3V3 and are 3.3 V logic: the master must not drive them above 3.6 V. Each breakout also has its own 10uF + 100nF on VIN, so no extra ADC decoupling is fitted here.
3. Input filters are optional: with the 0R links and no capacitor fitted there is none (original behaviour). Fit 1k in each link and 100nF across A0-A1 for about an 800 Hz low-pass (the 100nF positions are marked DNP).
4. ADS #2: bridge its ADDR0 solder jumper (ADDR0 is pulled to GND by 10k, so the jumper ties it to VCC: expected address 0x41; the main board's firmware scans at boot). The Adafruit breakouts carry 10k pull-ups on SDA/SCL to their VIN.
5. Use shielded twisted cable for the gauge wires, shield to GND at this board only.

HastaArduino	
Sheet: /	
File: gaugeboard.kicad_sch	
Title: HastaDAQ strain-gauge board	
Size: A3	Date:
KiCad E.D.A. 10.0.6	Rev: A
	Id: 1/1